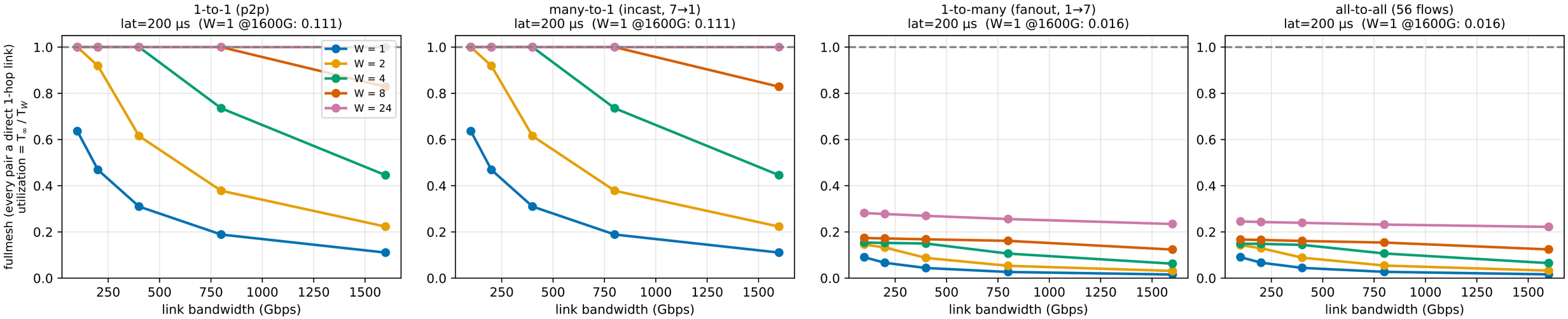
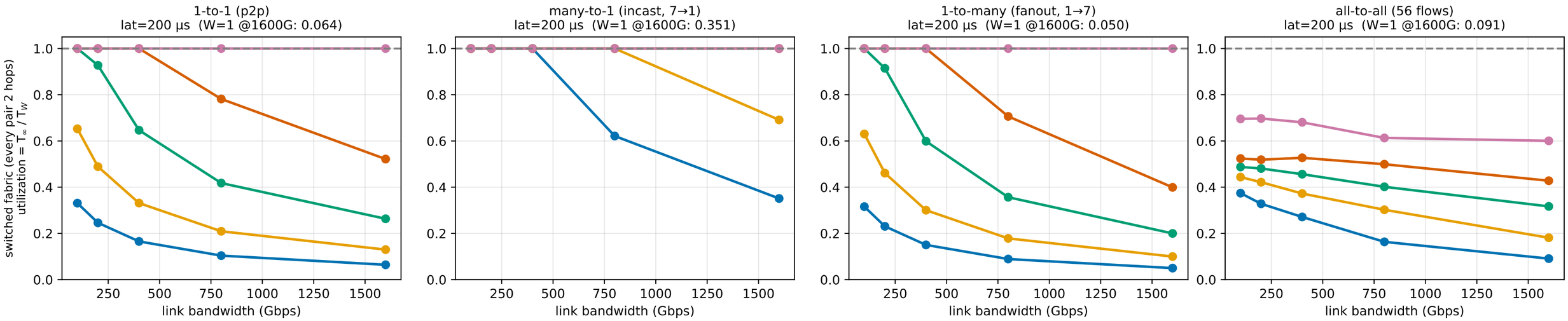
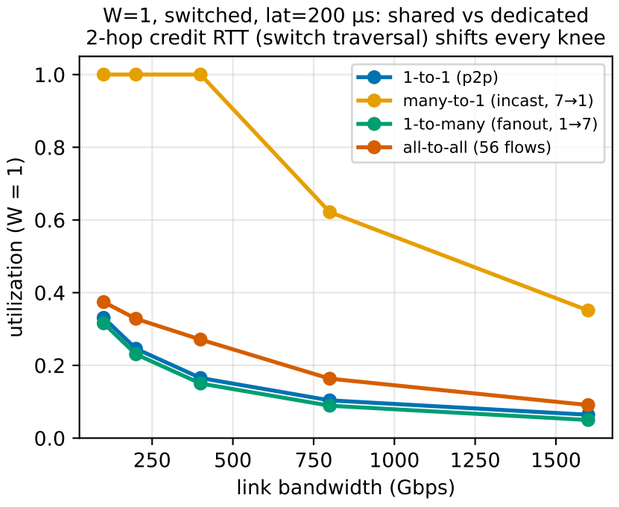
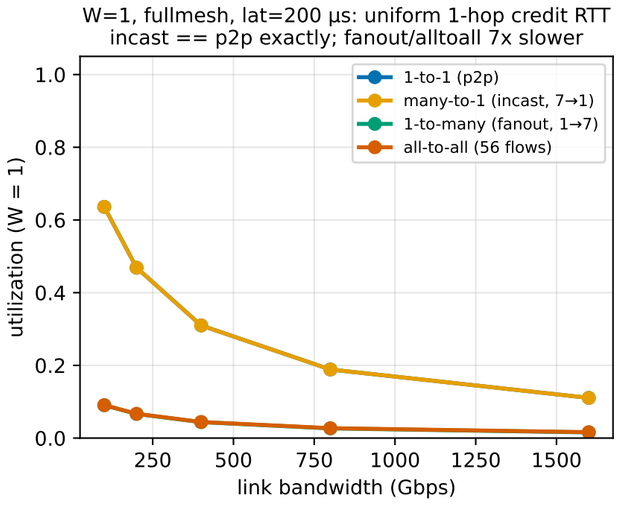
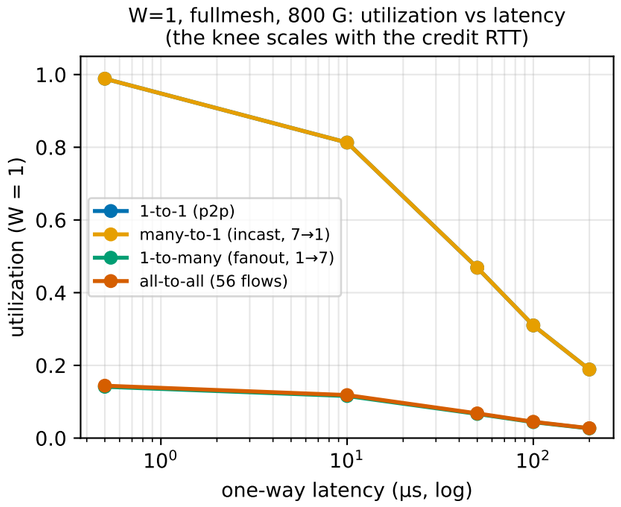
Credit flow control across traffic patterns on the Su fabric (8-GPU, 256 MiB per flow, fullmesh + switched):
window × chunk vs bandwidth-delay product — dedicated (p2p/incast) vs shared (fanout/alltoall) credit windows











Raw times @800 G, lat=200 µs, W=1 (ms)

fullmesh 1-to-1 (p2p)	15.28	(1.00x p2p)
fullmesh many-to-1 (incast, 7→1)	15.28	(1.00x p2p)
fullmesh 1-to-many (fanout, 1→7)	108.19	(7.08x p2p)
fullmesh all-to-all (56 flows)	108.26	(7.08x p2p)
switched 1-to-1 (p2p)	30.58	
switched many-to-1 (incast, 7→1)	31.09	
switched 1-to-many (fanout, 1→7)	216.47	
switched all-to-all (56 flows)	217.12	

BDP law per FLOW: W·chunk/(ser + 2·lat).
One VC per endpoint -> fanout and alltoall
share ONE window across their flows, so the
sender serializes 224 chunks through W:
W=1 fullmesh = 7.1x p2p (uniform 1-hop RTT;
the switched fabric adds a switch traversal
to every credit RTT and doubles the p2p
cycle).